Muhammad Bilal Shahid

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Summary

Postdoctoral researcher at Iowa State University working on uncertainty-aware machine learning for aerospace dynamical systems, including spacecraft and aircraft trajectory forecasting. Develops Predictor–Corrector frameworks that correct and calibrate learned and physics-based forecasters using Neural CDEs and Modern Hopfield Networks. PhD in Mechanical Engineering with a minor in Computer Engineering. Skilled in Python, PyTorch, JAX, and HPC workflows.

Education

PhD - **Iowa State University** Jan 2021 – May 2026

- **Major**: Mechanical Engineering **Minor**: Computer Engineering CGPA: 3.67/4.00

MS - **Incheon National University** Jan 2018 – Feb 2020

- **Major**: Mechanical Engineering

BS - **GIK Institute of Engineering Sciences & Technology** Aug 2012 – May 2016

- **Major**: Mechanical Engineering

Experience

Postdoc Research Associate - Iowa State University Jun 2026 – Present

- Developing calibrated uncertainty quantification for LLM-assisted STPA hazard analysis (ONR MURI)
- Developing calibrated, uncertainty-aware forecasting of aircraft trajectories at scale (RTX)
- Developed physics-as-architecture neural surrogates for structured dynamics (AIAA SciTech 2027)

PhD Researcher - Iowa State University Jan 2021 – May 2026

- Developed HopCast, a Modern Hopfield Network Corrector for calibrated autoregressive forecasts (TMLR 2025)
- Developed Neural CDE Correctors that reduce error accumulation in multi-step forecasts
- Developed a Latent Neural CDE Corrector for uncertainty-aware spacecraft trajectory forecasting with GMAT
- Developed deep-ensemble uncertainty quantification for cyber-physical-human systems (NASA STTR)
- Identified target variables for robust black-box car-following models (TRR 2025)
- Developed a hybrid neural-mechanistic crop model with uncertainty propagation

Graduate Research Assistant - Incheon National University Jan 2018 – Jul 2020

- Integrated machine learning with ultrasonic metal welding for process monitoring and design
- Developed a finite element and neural network approach to ultrasonic weld horn design
- Developed weld attributes for sound joints via nanoindentation and optical microscopy

Intern - Fauji Fertilizer Company May 2015 – Aug 2015

- Worked on the maintenance of boilers, heat exchangers, and cooling towers

Research Projects Supported by External Funding

Calibrated Multi-Agent Aircraft Trajectory Prediction Using Transformer Architectures - RTX Aug 2026 – Present

- Funded by Raytheon Technologies (RTX)
- Role: Postdoc Research Associate, developing calibrated multi-step forecasting of aircraft trajectories

Foundations of Dynamic Certification for Autonomy - ONR MURI May 2025 – Present

- Funded by the Office of Naval Research, Grant N00014-25-1-2479
- Role: Postdoc Research Associate, developing calibrated uncertainty quantification for LLM-assisted STPA

Uncertainty Quantification of Representations of Unknown Dynamic Systems through Universal Differential Equations - NASA STTR Jul 2022 – Jan 2026

- Funded by NASA through STTR awards to Elder Research, Inc., with Iowa State University as research institution
- Phase I (80NSSC22PB011), Jul 2022 – Aug 2023
- Phase II (80NSSC24CA037), Jan 2024 – Jan 2026
- Role: Graduate Researcher, developing deep-ensemble uncertainty quantification and trajectory correctors

Publications

Accepted

- **SHAHID, M. B.**, AND FLEMING, C. Physics-as-architecture for learning structured dynamics. In AIAA SciTech 2027 Forum (2027).
- **SHAHID, M. B.**, JIANG, Z., SARKAR, S., AND FLEMING, C. Probabilistic correction for GMAT-based spacecraft trajectory forecasting. In AIAA SciTech 2027 Forum (2027).

Under Review

- **SHAHID, M. B.**, JIANG, Z., SARKAR, S., AND FLEMING, C. Continuous-time probabilistic correctors for uncertainty-aware physics-based spacecraft trajectory forecasting. Aerospace Science and Technology (2026). arXiv:2606.21021.
- **SHAHID, M. B.**, JIANG, Z., KOIRALA, P., SARKAR, S., AND FLEMING, C. Neural CDEs as correctors for learned time series models. Neural Networks (2026). arXiv:2512.12116.
- SHARIATEE, M., **SHAHID, M. B.**, AND FLEMING, C. SKIP: A dynamic bipartite-graph matching policy for profit maximization in autonomous mobility-on-demand systems. In IEEE International Conference on Robotics and Automation (ICRA) (2027).

Published

- **SHAHID, M. B.**, AND FLEMING, C. HopCast: Calibration of autoregressive dynamics models. Transactions on Machine Learning Research (2025). openreview.net/forum?id=wsO6nxvGof.
- **SHAHID, M. B.**, AND FLEMING, C. Toward robust car-following dynamics modeling via blackbox models: Methodology, analysis, and recommendations. Transportation Research Record 2679, 4 (2025), 114–128. doi:10.1177/03611981241287540.
- **SHAHID, M. B.**, ROBISON, R., SHAFER, T., DILORETO, V., ALEXANDROV, N. M., AND FLEMING, C. Uncertainty quantification using deep ensembles for decision making in cyber-physical-human systems. In AIAA SciTech 2024 Forum (2024).
- ROBISON, R., SHAFER, T., DILORETO, V., ALEXANDROV, N. M., **SHAHID, M. B.**, AND FLEMING, C. Differential equation approximation using gradient-boosted quantile regression. In AIAA SciTech 2024 Forum (2024), p. 0107.
- **SHAHID, M. B.**, AND PARK, D.-S. In-process weld quality monitoring by the multi-layer perceptron neural network in ultrasonic metal welding. Journal of the Korean Society of Manufacturing Process Engineers 21, 6 (2022), 89–97. doi:10.14775/ksmpe.2022.21.06.089.
- **SHAHID, M. B.**, JUNG, J.-Y., AND PARK, D.-S. Finite element analysis coupled artificial neural network approach to design the longitudinal-torsional mode ultrasonic welding horn. The International Journal of Advanced Manufacturing Technology 107, 5 (2020), 2731–2743. doi:10.1007/s00170-020-05200-5.
- **SHAHID, M. B.**, HAN, S.-C., JUN, T.-S., AND PARK, D.-S. Effect of process parameters on the joint strength in ultrasonic welding of Cu and Ni foils. Materials and Manufacturing Processes 34, 11 (2019), 1217–1224. doi:10.1080/10426914.2019.1643474.
- **SHAHID, M. B.**, AND PARK, D.-S. Effect of welding parameters on joint strength in ultrasonic metal welding. In Proceedings of the Korean Society for Precision Engineering Conference (2019), p. 762.
- LEE, Y. J., **SHAHID, M. B.**, AND PARK, D.-S. Designing of ultrasonic horns to improve amplitude uniformity in ultrasonic metal welding. In MATEC Web of Conferences (2019), vol. 257, EDP Sciences, p. 02009. doi:10.1051/matecconf/201925702009.

Research Projects

Correction of Dynamics Models via Sequence Modeling Paper | Codebase

- Developed a Predictor–Corrector framework where a Neural CDE Corrector corrects a learned Predictor
- The Corrector handles irregular sampling and works with continuous- and discrete-time Predictors
- Introduced two regularization strategies that improve extrapolation and accelerate training
- **Tools Used**: Python, JAX, Diffrax

Calibration of Dynamics Models via Pattern Retrieval Paper | Codebase

- Developed a Predictor–Corrector framework where a Modern Hopfield Network Corrector retrieves residuals
- The retrieved residuals yield calibrated prediction intervals for autoregressive forecasts
- Published in Transactions on Machine Learning Research (2025)
- **Tools Used**: Python, PyTorch, Modern Hopfield Networks

Robust Car-Following Dynamics Modeling Paper | Codebase

- Studied target variable selection for training robust black-box car-following models
- Compared target variables for black-box models in the function and parameter spaces
- Published in Transportation Research Record (2025) with recommendations per model class
- **Tools Used**: Python, PyTorch, LSTM, Ridge Regression

Uncertainty-Aware Spacecraft Trajectory Forecasting Paper | Codebase

- Developed a Predictor–Corrector framework where a learned Corrector wraps NASA's GMAT propagator
- The Corrector is a Latent Neural CDE that models the evolution of forecast errors in continuous time
- The Corrector yields calibrated full-covariance uncertainty over 2–4 day horizons
- Evaluated on real-world ephemeris data from NASA's Crustal Dynamics Data Information System
- A related conference paper is accepted at AIAA SciTech 2027
- **Tools Used**: Python, JAX, Diffrax, GMAT

Achievements

- Silver Reviewer Award, ICML 2026, placing among the top reviewers based on area chair ratings
- Two first-author papers accepted at AIAA SciTech 2027, Orlando, Florida
- Awarded the International Student Scholarship for MS at Incheon National University, South Korea
- Second position in the Board of Intermediate Exam, Bahawalpur, Pakistan (11th & 12th grade)
- Certificate of Merit from the Chief Minister of Punjab for outstanding performance in the Intermediate Exam

Talks

- Uncertainty Quantification of Representations of Unknown Dynamic Systems through Machine Learning. Joint presentation with Elder Research, Autonomy Tag-up, NASA Langley Research Center, 12 Mar 2025

Academic Service

- **Conference Reviewer** (32 submissions): NeurIPS 2026 (4), ICML 2026 (5), ICLR 2027 (1), AAAI 2027 Program Committee (6), KDD 2026 & 2027 AI4Sciences Track (6), AIAA SciTech 2027 (10)
- **Journal Reviewer**: Journal of Aerospace Information Systems (2026)
- Mentored REU (Research Experience for Undergraduates) interns, Summer 2022 & Summer 2026
- Received a Certificate of Appreciation for exemplary leadership and mentorship of REU interns

Skills

Programming: Python, C++, GitHub, LaTeX, Linux, JAX, PyTorch, NumPy, Matplotlib, MATLAB

Deep Learning: Supervised Learning, Unsupervised Learning, Uncertainty Quantification, Reinforcement Learning

Time Series: Time Series Forecasting, Differential Equations, Autoregressive Models, Pattern Retrieval